

Modelling, Uncertainty and Data for Engineers

Assignment 1

CHITRITA BARMAN (CE26D015)

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Problem Statement

For the function

$$f(x) = \sin x,$$

the exact derivative is

$$f'(x) = \cos x.$$

We evaluate the numerical derivative at three points

$$x_1 = \frac{\pi}{4}, \quad x_2 = \frac{\pi}{2}, \quad x_3 = \pi,$$

using the forward, backward, and central finite-difference formulas with a fixed step size

$$\boxed{h = 0.01}.$$

The finite-difference formulas are:

$$\text{Forward difference: } f'_F(x) = \frac{f(x+h) - f(x)}{h}, \quad (1)$$

$$\text{Backward difference: } f'_B(x) = \frac{f(x) - f(x-h)}{h}, \quad (2)$$

$$\text{Central difference: } f'_C(x) = \frac{f(x+h) - f(x-h)}{2h}. \quad (3)$$

The absolute error for each method is defined as

$$E_{\text{abs}} = |f'_{\text{numerical}}(x) - f'_{\text{exact}}(x)|.$$

1. Calculations at $x = \frac{\pi}{4}$

Exact derivative

$$f'_{\text{exact}}\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} = 0.7071067812.$$

Function values

$$\begin{aligned}f\left(\frac{\pi}{4}\right) &= \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} = 0.7071067812, \\f\left(\frac{\pi}{4} + h\right) &= \sin\left(\frac{\pi}{4} + 0.01\right) = 0.7141423761, \\f\left(\frac{\pi}{4} - h\right) &= \sin\left(\frac{\pi}{4} - 0.01\right) = 0.7000004762.\end{aligned}$$

Forward difference

Using Eq. (1):

$$\begin{aligned}f'_F\left(\frac{\pi}{4}\right) &= \frac{\sin\left(\frac{\pi}{4} + 0.01\right) - \sin\left(\frac{\pi}{4}\right)}{0.01} \\&= \frac{0.7141423761 - 0.7071067812}{0.01} \\&= \frac{0.0070355949}{0.01} \\&= 0.70355949.\end{aligned}$$

Absolute error:

$$\begin{aligned}E_F &= |0.70355949 - 0.7071067812| \\&= 0.00354729.\end{aligned}$$

Backward difference

Using Eq. (2):

$$\begin{aligned}f'_B\left(\frac{\pi}{4}\right) &= \frac{\sin\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4} - 0.01\right)}{0.01} \\&= \frac{0.7071067812 - 0.7000004762}{0.01} \\&= \frac{0.007106305}{0.01} \\&= 0.7106305.\end{aligned}$$

Absolute error:

$$\begin{aligned}E_B &= |0.7106305 - 0.7071067812| \\&= 0.00352372.\end{aligned}$$

Central difference

Using Eq. (3):

$$\begin{aligned}f'_C\left(\frac{\pi}{4}\right) &= \frac{\sin\left(\frac{\pi}{4} + 0.01\right) - \sin\left(\frac{\pi}{4} - 0.01\right)}{2(0.01)} \\&= \frac{0.7141423761 - 0.7000002762}{0.02} \\&= \frac{0.0141420999}{0.02} \\&= 0.70709500.\end{aligned}$$

Absolute error:

$$\begin{aligned}E_C &= |0.70709500 - 0.7071067812| \\&= 0.00001178.\end{aligned}$$

2. Calculations at $x = \frac{\pi}{2}$

Exact derivative

$$f'_{\text{exact}}\left(\frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2}\right) = 0.$$

Function values

$$\begin{aligned}f\left(\frac{\pi}{2}\right) &= \sin\left(\frac{\pi}{2}\right) = 1, \\f\left(\frac{\pi}{2} + h\right) &= \sin\left(\frac{\pi}{2} + 0.01\right) = \cos(0.01) = 0.9999500004, \\f\left(\frac{\pi}{2} - h\right) &= \sin\left(\frac{\pi}{2} - 0.01\right) = \cos(0.01) = 0.9999500004.\end{aligned}$$

Forward difference

Using Eq. (1):

$$\begin{aligned}f'_F\left(\frac{\pi}{2}\right) &= \frac{\sin\left(\frac{\pi}{2} + 0.01\right) - \sin\left(\frac{\pi}{2}\right)}{0.01} \\&= \frac{0.9999500004 - 1}{0.01} \\&= \frac{-0.0000499996}{0.01} \\&= -0.00499996.\end{aligned}$$

Absolute error:

$$\begin{aligned}E_F &= |-0.00499996 - 0| \\&= 0.00499996.\end{aligned}$$

Backward difference

Using Eq. (2):

$$\begin{aligned}f'_B\left(\frac{\pi}{2}\right) &= \frac{\sin\left(\frac{\pi}{2}\right) - \sin\left(\frac{\pi}{2} - 0.01\right)}{0.01} \\&= \frac{1 - 0.9999500004}{0.01} \\&= \frac{0.0000499996}{0.01} \\&= 0.00499996.\end{aligned}$$

Absolute error:

$$\begin{aligned}E_B &= |0.00499996 - 0| \\&= 0.00499996.\end{aligned}$$

Central difference

Using Eq. (3):

$$\begin{aligned}f'_C\left(\frac{\pi}{2}\right) &= \frac{\sin\left(\frac{\pi}{2} + 0.01\right) - \sin\left(\frac{\pi}{2} - 0.01\right)}{2(0.01)} \\&= \frac{0.9999500004 - 0.9999500004}{0.02} \\&= \frac{0}{0.02} \\&= 0.\end{aligned}$$

Absolute error:

$$E_C = |0 - 0| = 0.$$

3. Calculations at $x = \pi$

Exact derivative

$$f'_{\text{exact}}(\pi) = \cos(\pi) = -1.$$

Function values

$$\begin{aligned}f(\pi) &= \sin(\pi) = 0, \\f(\pi + h) &= \sin(\pi + 0.01) = -\sin(0.01) = -0.0099998333, \\f(\pi - h) &= \sin(\pi - 0.01) = \sin(0.01) = 0.0099998333.\end{aligned}$$

Forward difference

Using Eq. (1):

$$\begin{aligned}f'_F(\pi) &= \frac{\sin(\pi + 0.01) - \sin(\pi)}{0.01} \\&= \frac{-0.0099998333 - 0}{0.01} \\&= \frac{-0.0099998333}{0.01} \\&= -0.99998333.\end{aligned}$$

Absolute error:

$$\begin{aligned}E_F &= |-0.99998333 - (-1)| \\&= |-0.99998333 + 1| \\&= 0.00001667.\end{aligned}$$

Backward difference

Using Eq. (2):

$$\begin{aligned}f'_B(\pi) &= \frac{\sin(\pi) - \sin(\pi - 0.01)}{0.01} \\&= \frac{0 - 0.0099998333}{0.01} \\&= \frac{-0.0099998333}{0.01} \\&= -0.99998333.\end{aligned}$$

Absolute error:

$$\begin{aligned}E_B &= |-0.99998333 - (-1)| \\&= 0.00001667.\end{aligned}$$

Central difference

Using Eq. (3):

$$\begin{aligned}f'_C(\pi) &= \frac{\sin(\pi + 0.01) - \sin(\pi - 0.01)}{2(0.01)} \\&= \frac{-0.0099998333 - 0.0099998333}{0.02} \\&= \frac{-0.0199996666}{0.02} \\&= -0.99998333.\end{aligned}$$

Absolute error:

$$\begin{aligned}E_C &= |-0.99998333 - (-1)| \\&= 0.00001667.\end{aligned}$$

Summary of results for $h = 0.01$

Point x	Method	Approx. derivative	Exact derivative	Absolute error
$\frac{\pi}{4}$	Forward	0.70355949	0.70710678	0.00354729
	Backward	0.7106305	0.70710678	0.00352372
	Central	0.70709500	0.70710678	0.00001178
$\frac{\pi}{2}$	Forward	-0.00499996	0	0.00499996
	Backward	0.00499996	0	0.00499996
	Central	0	0	0
π	Forward	-0.99998333	-1	0.00001667
	Backward	-0.99998333	-1	0.00001667
	Central	-0.99998333	-1	0.00001667

Let's take a new step size

$$\boxed{h = 0.1}.$$

1. Calculations at $x = \frac{\pi}{4}$

Exact derivative

$$f'_{\text{exact}}\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} = 0.7071067812.$$

Function values

$$\begin{aligned} f\left(\frac{\pi}{4}\right) &= \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} = 0.7071067812, \\ f\left(\frac{\pi}{4} + h\right) &= \sin\left(\frac{\pi}{4} + 0.1\right) = 0.7741670785, \\ f\left(\frac{\pi}{4} - h\right) &= \sin\left(\frac{\pi}{4} - 0.1\right) = 0.6329813067. \end{aligned}$$

Forward difference

Using Eq. (1):

$$\begin{aligned}
f'_F\left(\frac{\pi}{4}\right) &= \frac{\sin\left(\frac{\pi}{4} + 0.1\right) - \sin\left(\frac{\pi}{4}\right)}{0.1} \\
&= \frac{0.7741670785 - 0.7071067812}{0.1} \\
&= \frac{0.0670602973}{0.1} \\
&= 0.6706029729.
\end{aligned}$$

Absolute error:

$$\begin{aligned}
E_F &= |0.6706029729 - 0.7071067812| \\
&= 0.0365038083.
\end{aligned}$$

Backward difference

Using Eq. (2):

$$\begin{aligned}
f'_B\left(\frac{\pi}{4}\right) &= \frac{\sin\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4} - 0.1\right)}{0.1} \\
&= \frac{0.7071067812 - 0.6329813067}{0.1} \\
&= \frac{0.0741254745}{0.1} \\
&= 0.7412547451.
\end{aligned}$$

Absolute error:

$$\begin{aligned}
E_B &= |0.7412547451 - 0.7071067812| \\
&= 0.0341479639.
\end{aligned}$$

Central difference

Using Eq. (3):

$$\begin{aligned}
f'_C\left(\frac{\pi}{4}\right) &= \frac{\sin\left(\frac{\pi}{4} + 0.1\right) - \sin\left(\frac{\pi}{4} - 0.1\right)}{2(0.1)} \\
&= \frac{0.7741670785 - 0.6329813067}{0.2} \\
&= \frac{0.1411857718}{0.2} \\
&= 0.7059288590.
\end{aligned}$$

Absolute error:

$$\begin{aligned} E_C &= |0.7059288590 - 0.7071067812| \\ &= 0.0011779222. \end{aligned}$$

2. Calculations at $x = \frac{\pi}{2}$

Exact derivative

$$f'_{\text{exact}}\left(\frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2}\right) = 0.$$

Function values

$$\begin{aligned} f\left(\frac{\pi}{2}\right) &= \sin\left(\frac{\pi}{2}\right) = 1, \\ f\left(\frac{\pi}{2} + h\right) &= \sin\left(\frac{\pi}{2} + 0.1\right) = \cos(0.1) = 0.9950041653, \\ f\left(\frac{\pi}{2} - h\right) &= \sin\left(\frac{\pi}{2} - 0.1\right) = \cos(0.1) = 0.9950041653. \end{aligned}$$

Forward difference

Using Eq. (1):

$$\begin{aligned} f'_F\left(\frac{\pi}{2}\right) &= \frac{\sin\left(\frac{\pi}{2} + 0.1\right) - \sin\left(\frac{\pi}{2}\right)}{0.1} \\ &= \frac{0.9950041653 - 1}{0.1} \\ &= \frac{-0.0049958347}{0.1} \\ &= -0.0499583472. \end{aligned}$$

Absolute error:

$$\begin{aligned} E_F &= |-0.0499583472 - 0| \\ &= 0.0499583472. \end{aligned}$$

Backward difference

Using Eq. (2):

$$\begin{aligned}
f'_B\left(\frac{\pi}{2}\right) &= \frac{\sin\left(\frac{\pi}{2}\right) - \sin\left(\frac{\pi}{2} - 0.1\right)}{0.1} \\
&= \frac{1 - 0.9950041653}{0.1} \\
&= \frac{0.0049958347}{0.1} \\
&= 0.0499583472.
\end{aligned}$$

Absolute error:

$$\begin{aligned}
E_B &= |0.0499583472 - 0| \\
&= 0.0499583472.
\end{aligned}$$

Central difference

Using Eq. (3):

$$\begin{aligned}
f'_C\left(\frac{\pi}{2}\right) &= \frac{\sin\left(\frac{\pi}{2} + 0.1\right) - \sin\left(\frac{\pi}{2} - 0.1\right)}{2(0.1)} \\
&= \frac{0.9950041653 - 0.9950041653}{0.2} \\
&= 0.
\end{aligned}$$

Absolute error:

$$E_C = |0 - 0| = 0.$$

3. Calculations at $x = \pi$

Exact derivative

$$f'_{\text{exact}}(\pi) = \cos(\pi) = -1.$$

Function values

$$\begin{aligned}
f(\pi) &= \sin(\pi) = 0, \\
f(\pi + h) &= \sin(\pi + 0.1) = -\sin(0.1) = -0.0998334166, \\
f(\pi - h) &= \sin(\pi - 0.1) = \sin(0.1) = 0.0998334166.
\end{aligned}$$

Forward difference

Using Eq. (1):

$$\begin{aligned}f'_F(\pi) &= \frac{\sin(\pi + 0.1) - \sin(\pi)}{0.1} \\&= \frac{-0.0998334166 - 0}{0.1} \\&= -0.9983341665.\end{aligned}$$

Absolute error:

$$\begin{aligned}E_F &= |-0.9983341665 - (-1)| \\&= 0.0016658335.\end{aligned}$$

Backward difference

Using Eq. (2):

$$\begin{aligned}f'_B(\pi) &= \frac{\sin(\pi) - \sin(\pi - 0.1)}{0.1} \\&= \frac{0 - 0.0998334166}{0.1} \\&= -0.9983341665.\end{aligned}$$

Absolute error:

$$\begin{aligned}E_B &= |-0.9983341665 - (-1)| \\&= 0.0016658335.\end{aligned}$$

Central difference

Using Eq. (3):

$$\begin{aligned}f'_C(\pi) &= \frac{\sin(\pi + 0.1) - \sin(\pi - 0.1)}{2(0.1)} \\&= \frac{-0.0998334166 - 0.0998334166}{0.2} \\&= \frac{-0.1996668332}{0.2} \\&= -0.9983341665.\end{aligned}$$

Absolute error:

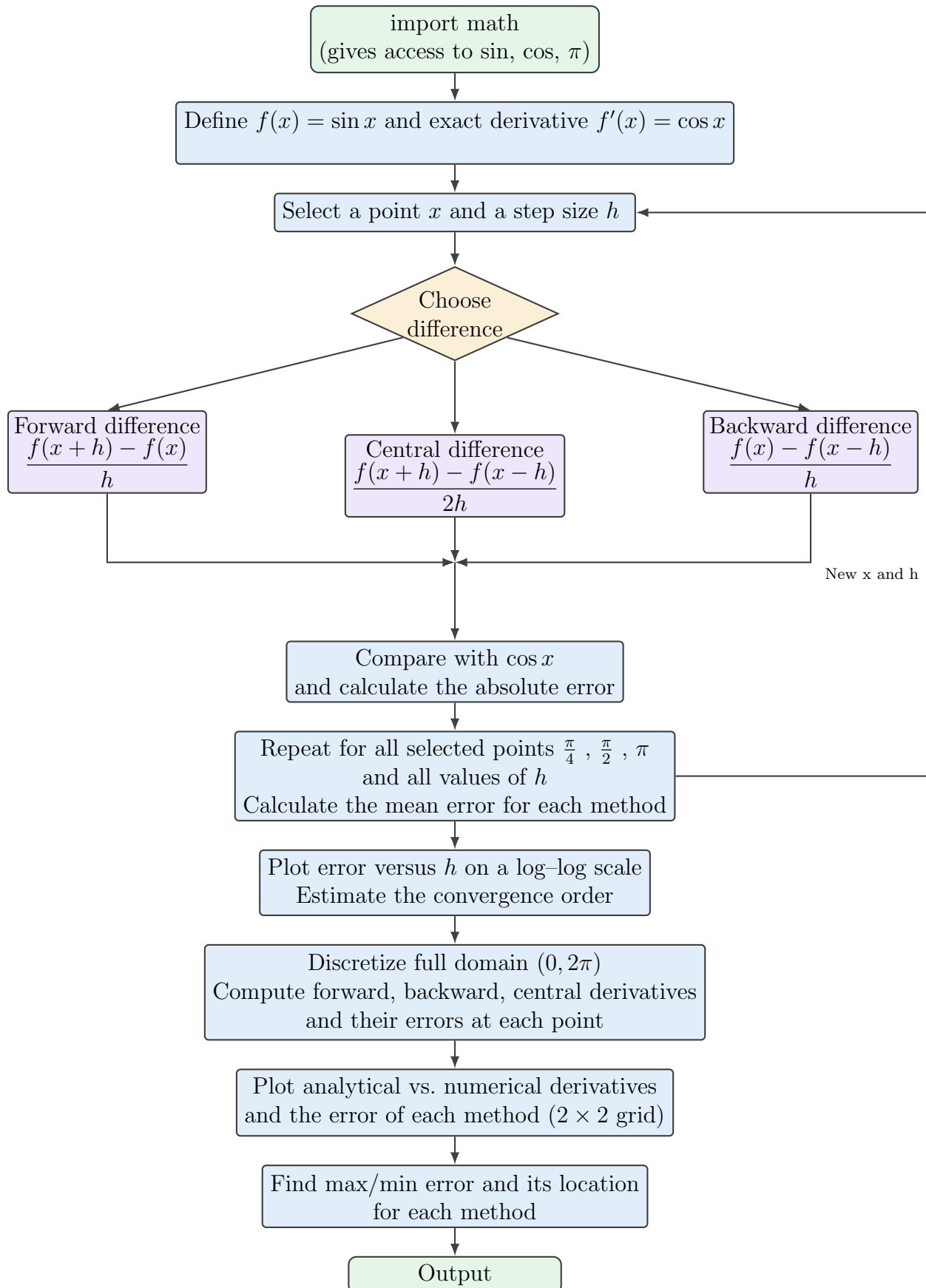
$$\begin{aligned}E_C &= |-0.9983341665 - (-1)| \\&= 0.0016658335.\end{aligned}$$

Summary of results for $h = 0.1$

Point x	Method	Approx. derivative	Exact derivative	Absolute error
$\frac{\pi}{4}$	Forward	0.67060297	0.70710678	0.03650381
	Backward	0.74125475	0.70710678	0.03414796
	Central	0.70592886	0.70710678	0.00117792
$\frac{\pi}{2}$	Forward	-0.04995835	0	0.04995835
	Backward	0.04995835	0	0.04995835
	Central	0	0	0
π	Forward	-0.99833417	-1	0.00166583
	Backward	-0.99833417	-1	0.00166583
	Central	-0.99833417	-1	0.00166583

Finite Difference Method Flow Chart

Numerical derivative of $f(x) = \sin x$



Newton–Raphson Method Flow Chart

Roots of $f(x) = \sin x$ in $(0, 4\pi)$

